

Worksheet 2
MTH 203 (Multivariate Calculus)
Date: 25-08-2026

Q1. Find the interior, exterior and boundary of the sets

- (a) $S_1 = \{x \in \mathbb{R}^n : \|x\| \leq 1\};$
- (b) $S_2 = \{x \in \mathbb{R}^n : \|x\| = 1\};$
- (c) $S_3 = \{x \in \mathbb{R}^n : x_i \text{ is rational}\}.$

Q2. Prove that the union of any (even infinite) number of open sets is open. Prove that the intersection of two (and hence of finitely many) open sets is open. Give a counterexample for infinitely many open sets.

Q3. Let S be a subset of $\mathbb{R}^n$. Prove that $\text{int}(S)$ is open subset of $\mathbb{R}^n$.

Q4. Let S be the set of all points $(x, y) \in \mathbb{R}^n$ satisfying the given conditions. Make a sketch showing the set S and give a geometric argument to explain whether S is open, closed, both open and closed, or neither open nor closed.

- (a) $y = x^2;$
- (b) $1 < x^2 + y^2 \leq 2.;$
- (c) $1 \leq x \leq 2, 3 \leq y \leq 4..$

Q5. Find the limit of f as $(x, y) \rightarrow (0, 0)$ or show that the limit does not exist (using polar coordinate)

- (a) $f(x, y) = \cos\left(\frac{x^3 - y^3}{x^2 + y^2}\right);$
- (b) $f(x, y) = \tan^{-1}\left(\frac{|x| + |y|}{x^2 + y^2}\right);$
- (c) $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$

Q6. Determine the largest set on which the given functions are continuous

- (a) $f(x, y) = \frac{1}{x^2 - y};$
- (b) $f(x, y, z) = \sqrt{x + y + z};$
- (c) $f(x, y, z) = \tan^{-1}\left(\frac{x + y}{1 - xy}\right)$
- (d) $f(x, y, z) = \frac{1}{|xy| + |z|}.$

Q7. Define $f(0, 0)$ in a way that extends $f(x, y) = xy \frac{x^2 - y^2}{x^2 + y^2}$ to be continuous at the origin.

Q8. In each example below determine the limit

- (a) $\lim_{(x,y) \rightarrow (0,0)} e^{y \frac{\sin x}{x}};$
- (b) $\lim_{(x,y) \rightarrow (\pi/2,0)} \frac{\cos y + 1}{y - \sin x};$
- (c) $\lim_{(x,y) \rightarrow (1,1)} \ln |1 + x^2 y^2|;$
- (d) $\lim_{(x,y) \rightarrow (0, \ln 2)} e^{x-y}.$

Bonus Questions (No marks)

- Q1.** Let $f(x, y) = 0$ if $y \leq 0$ or if $y \geq x^2$ and let $f(x, y) = 1$ if $0 < y < x^2$. Show that $f(x, y) \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$ along any straight line through the origin.
- Q2.** In the previous problem (1), find a curve through the origin along which (except at the origin) $f(x, y)$ has the constant-value 1. Is f continuous at the origin?